

SIMULATION BASED ANALYSIS ON PROMOTIONAL STRATEGIES IN RETAIL SALES

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March 2026

Abstract

This study presents a simulation-based analysis of promotional strategies in retail sales using machine learning techniques. A Random Forest regression model was developed to predict sales under various promotional conditions. Exploratory Data Analysis (EDA) was performed to understand sales patterns, promotional effects, and store characteristics. The model achieved an RMSE of approximately 1406 and an R^2 value of 0.796, indicating strong predictive performance. Seven promotional scenarios were simulated, including baseline, regular promotion, strong promotion, selective weekend promotion, store-type-based promotion, reduced promotion, and assortment-based promotion. Results show that strong promotions produce the highest uplift (58.25%), while targeted strategies achieve performance comparable to uniform promotion. The findings demonstrate that simulation modeling can effectively support data-driven retail decision-making.

1 Introduction

Retail performance is significantly influenced by promotional strategies. Businesses rely on promotions to increase customer traffic and revenue; however, determining the optimal promotional intensity and targeting approach remains challenging. This study integrates machine learning and simulation techniques to evaluate the impact of multiple promotional strategies and identify effective alternatives.

2 Literature Review

Sales promotion has long been recognized as a major driver of short-term retail performance. Blattberg and Neslin [1] demonstrated that promotional activities significantly influence consumer purchasing behavior and generate measurable sales uplift. Similarly, Neslin [2] showed that promotions affect purchase timing and brand-switching patterns, highlighting their strategic importance in retail environments. Store-level studies such as Van Heerde et al. [3] further revealed that the promotional “sales bump” varies across stores and conditions, supporting the need for targeted rather than uniform strategies.

With the growth of data-driven analytics, machine learning models have increasingly been used for retail sales forecasting. Breiman [4] introduced the Random Forest algorithm, which is effective in modeling complex nonlinear relationships in high-variability datasets. Extending predictive modeling toward strategic decision-making, Bertsimas

and Kallus [5] emphasized the role of prescriptive analytics in guiding optimal business strategies. Building on these foundations, the present study integrates predictive modeling with simulation-based analysis to evaluate alternative promotional strategies in a structured “what-if” framework.

3 Methodology

3.1 Data Preparation

The dataset was preprocessed using the following steps:

- Sales data and store data were merged.
- Closed store records were removed.
- Missing values were handled appropriately.
- Categorical variables (StoreType, Assortment) were encoded.
- Time-based features such as Month and Week were extracted.

3.2 Exploratory Data Analysis

EDA was performed to understand the relationships between promotions, store characteristics, customer count, and sales.

3.2.1 Promotion vs No Promotion

Average sales were significantly higher on promotion days compared to non-promotion days, confirming the strong positive impact of promotional activities.

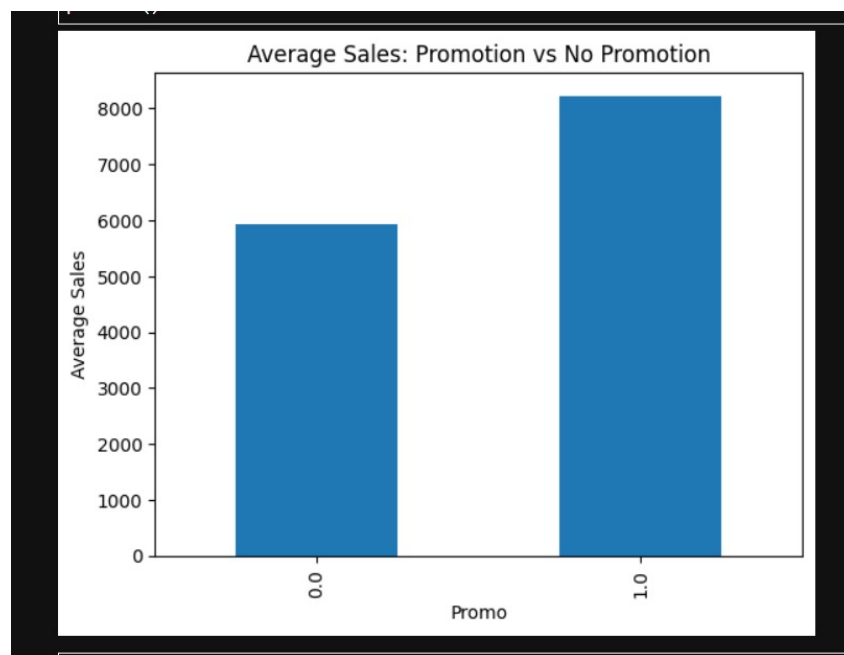


Figure 1: Average Sales: Promotion vs No Promotion

3.2.2 Distribution of Sales

Sales distribution was positively skewed, with most observations concentrated within a moderate range and a few extreme high-demand days.

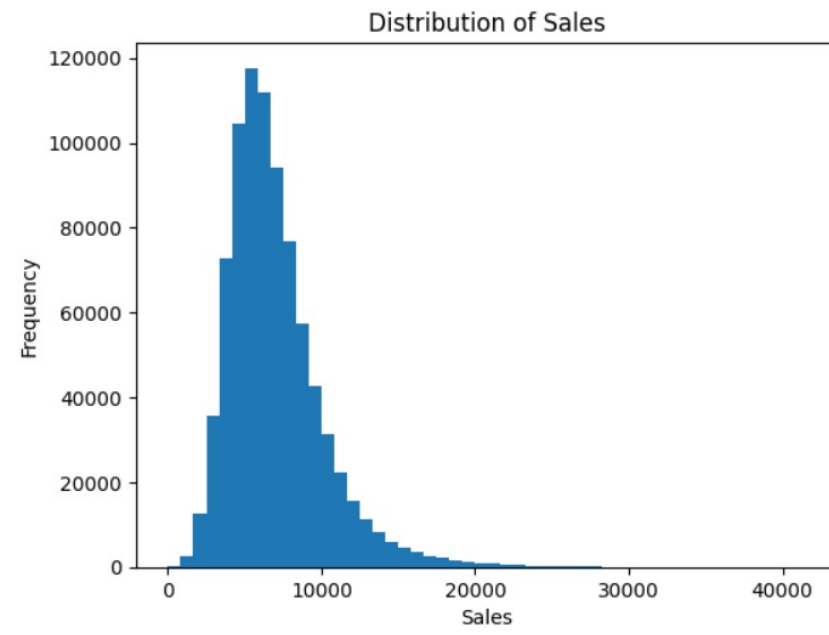


Figure 2: Distribution of Daily Sales

3.2.3 Average Sales by Day of Week

Significant variation was observed across weekdays, indicating demand seasonality.

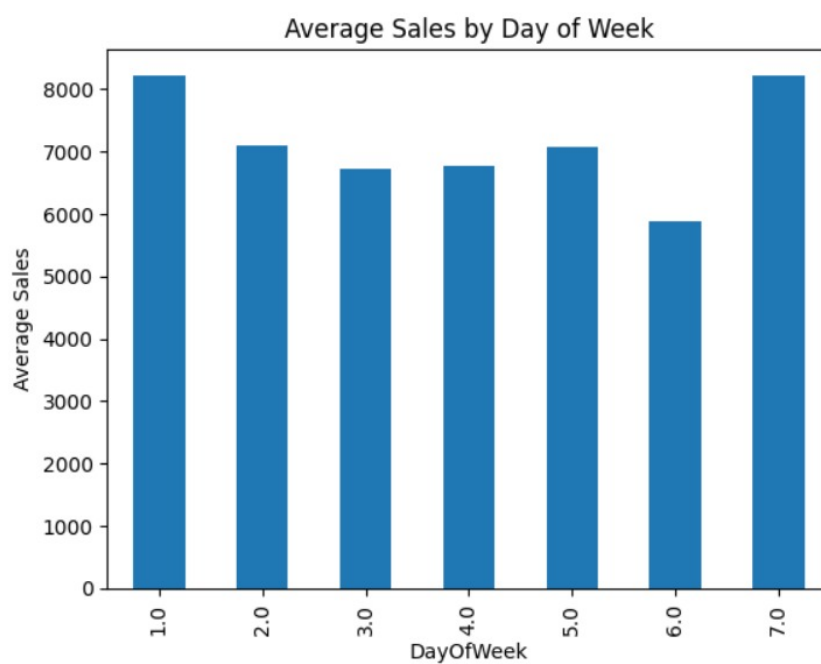


Figure 3: Average Sales by Day of Week

3.2.4 Average Sales by Assortment Type

Stores with expanded assortment (Type B) recorded significantly higher sales compared to basic assortment stores.

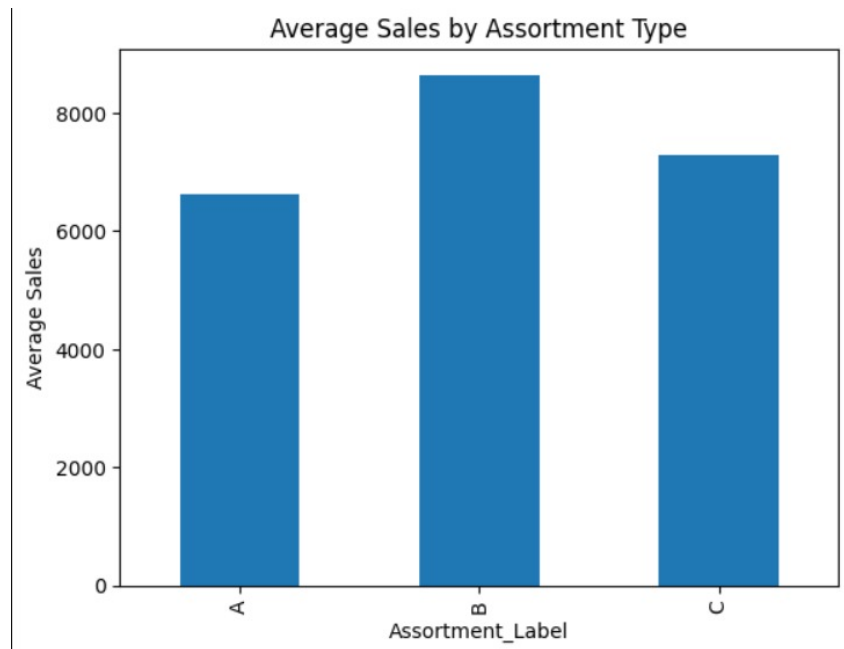


Figure 4: Average Sales by Assortment Type

3.2.5 Correlation Analysis

The correlation analysis reveals a strong positive relationship between sales and customer count (0.82), indicating that footfall is a primary driver of revenue. Promotions show a moderate positive correlation with sales (0.37), confirming their effectiveness while also suggesting that other demand factors influence performance. The weak correlation between competition distance and sales indicates potential non-linear effects, justifying the use of a tree-based model such as Random Forest.

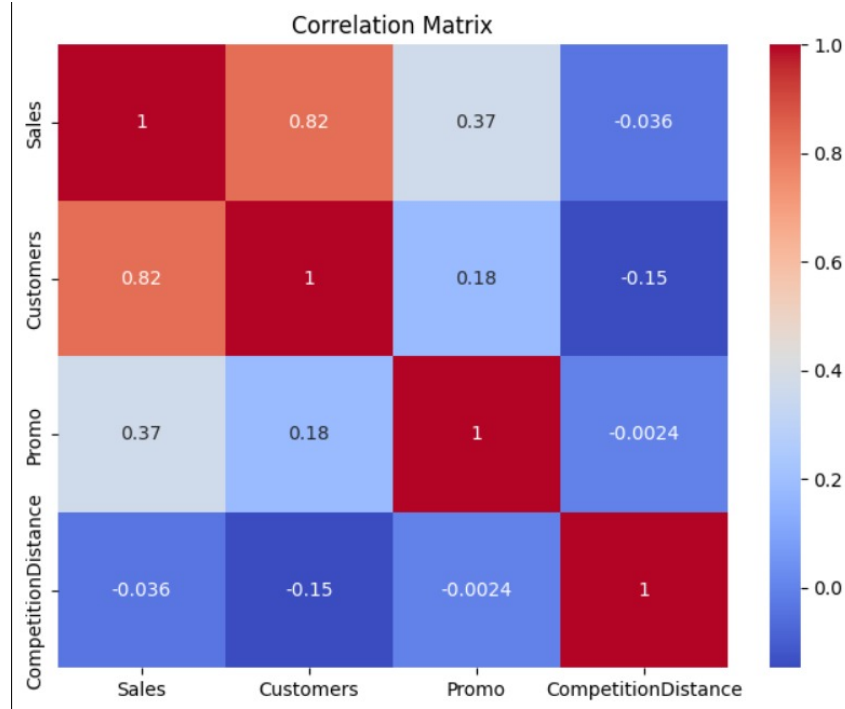


Figure 5: Correlation Matrix

3.2.6 Two-Sample t-Test

The 2-sample t-test was conducted to compare average sales between promotion and non-promotion days.

- Null Hypothesis: There is no significant difference in average sales between promotion and non-promotion days.
- Alternate Hypothesis: Average sales on promotional days are significantly higher than on non-promotion days.

The results show a statistically significant difference ($t=356.63$, $p < 0.05$), confirming that promotional activities have a significant positive impact on retail sales.

3.3 Predictive Model Development

A Random Forest regression model was developed.

Performance metrics:

$$RMSE \approx 1406$$

$$R^2 = 0.796$$

The model explains approximately 79.6% of sales variability.

3.4 Simulation Design

The simulation framework was developed using the trained Random Forest regression model to evaluate multiple hypothetical promotional strategies under controlled conditions. Each scenario was constructed by modifying key promotional variables (such as promotion intensity, timing, store-type targeting, and assortment-based targeting) while keeping other relevant store and temporal features consistent. The model was then used to generate predicted sales values for each modified dataset, allowing direct comparison against the baseline (no-promotion) scenario.

This structured “what-if” approach enabled the assessment of incremental sales uplift under different strategic conditions without conducting real-world experiments. By systematically altering promotional inputs and observing predicted outcomes, the simulation design provided a practical and risk-free mechanism for evaluating the effectiveness of alternative marketing strategies.

4 Results

4.1 Scenario 1: Baseline (No Promotion)

Predicted Sales: 4,994.99 million

- Scenario description: This scenario represents a baseline condition where no promotional activities are applied across stores. It serves as the reference scenario for evaluating all other promotional strategies.
- Key variables involved: Promo = 0, store characteristics, time-based features (Month, Week), sales.
- Conclusion: The baseline scenario establishes a benchmark against which all promotional strategies are compared. It highlights the level of revenue achievable without promotional support.

4.2 Scenario 2: Regular Promotion

Predicted Sales: 5,873.80 million Increase: 17.59%

- Scenario description: This scenario simulates the application of regular promotional activities across all stores and time periods.
- Key variables involved: Promo = 1, store attributes, temporal features, and sales history.
- Conclusion: Regular promotions significantly improve sales performance. This scenario confirms the strong positive effect of promotions on retail sales.

4.3 Scenario 3: Strong Promotion

Predicted Sales: 7,904.75 million Increase: 58.25%

- Scenario description: This scenario models an intensified promotional strategy where promotional influence is increased beyond regular levels.
- Key variables involved: Enhanced promotion indicator (Promo =1), store-level features, and time-based variables.

- Conclusion: Strong promotions lead to a substantial increase in predicted sales. However, such aggressive strategies may involve higher costs and should be evaluated carefully in practice.

4.4 Scenario 4: Weekend Promotion

Predicted Sales: 5,268.56 million Increase: 5.48%

- Scenario description: This scenario applies promotions selectively, restricting them primarily to weekend periods instead of continuous promotion.
- Key variables involved: DayOfWeek, along with store and time features.
- Conclusion: Selective weekend promotions result in a modest sales improvement. This suggests that limited promotional exposure reduces overall uplift compared to broader strategies.

4.5 Scenario 5: Store-Type-Based Promotion

Predicted Sales: 5,831.59 million Increase: 16.75%

- Scenario description: This scenario simulates promotions targeted only at specific store types, rather than applying promotions uniformly.
- Key variables involved: Promotion indicator conditioned on StoreType, along with temporal and store-specific attributes.
- Conclusion: Store-type-based promotions achieve sales gains comparable to regular promotions. This indicates that targeted strategies can be nearly as effective as uniform promotion.

4.6 Scenario 6: Reduced Promotion (50%)

Predicted Sales: 5,434.28 million Increase: 8.79%

- Scenario description: This scenario evaluates the effect of reducing promotional intensity by approximately 50% compared to the regular promotion scenario.
- Key variables involved: Reduced promotion indicator, store features, and temporal variables.
- Conclusion: Reducing promotional intensity lowers overall sales performance. This scenario highlights the trade-off between promotion strength and sales uplift.

4.7 Scenario 7: Assortment-Based Promotion

Predicted Sales: 5,871.99 million Increase: 17.56%

- Scenario description: This scenario applies promotional strategies selectively based on product assortment categories.
- Key variables involved: Promotion indicator conditioned on Assortment, along with store and time-related features.
- Conclusion: Assortment-based promotions achieve sales outcomes comparable to regular promotions. This suggests that targeted assortment-level strategies can be highly effective.

5 Discussion

The results indicate that promotional intensity significantly influences sales performance. Strong promotions generate maximum uplift but may involve higher costs. Targeted strategies such as store-type and assortment-based promotions provide nearly equivalent performance to regular promotion, suggesting improved efficiency. EDA findings support the nonlinear relationships captured by the Random Forest model.

6 Conclusion

This study presented a simulation-based evaluation of promotional strategies in retail sales using a Random Forest predictive model. The results indicate that promotional activities significantly increase sales compared to the baseline scenario. While strong promotions generated the highest uplift, targeted strategies such as store-type-based and assortment-based promotions achieved nearly comparable results, suggesting that intelligent targeting can be as effective as uniform promotional campaigns.

From a business perspective, the findings highlight the importance of data-driven decision-making in retail marketing. The integration of Exploratory Data Analysis and machine learning enabled a deeper understanding of demand patterns, customer influence, and promotional impact. Although the model performed strongly ($R^2 = 0.796$), challenges such as sales variability and external demand factors remain important considerations.

Future enhancements may include incorporating cost analysis to evaluate profitability instead of revenue alone, integrating additional external variables, and experimenting with advanced time-series models. Overall, the study demonstrates that simulation-based predictive analytics can serve as a practical decision-support tool for optimizing promotional strategies in retail environments.

7 Recommendations

- Implement targeted promotional strategies for optimal cost-performance balance.
- Use simulation tools for strategic marketing planning.
- Incorporate cost analysis in future research.

References

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